Dysferlinopathy (MONDO:0016145): A Comprehensive Disease Characteristics Report

Causal gene: *DYSF* (dysferlin), chromosome 2p13.2 | **Inheritance:** Autosomal recessive | **Category:** Mendelian

Summary

Dysferlinopathy is an autosomal recessive muscular dystrophy caused by biallelic loss-of-function variants in *DYSF*, the gene encoding dysferlin — a large (~237 kDa), Ca^{2+}-sensing, multi-C2-domain transmembrane protein responsible for repairing injured skeletal muscle plasma membrane (sarcolemma). When both *DYSF* alleles are defective, muscle fibers lose the capacity to reseal the micro-tears generated by normal contraction. The consequence is a self-amplifying cycle of membrane leakage, complement activation, inflammation, myofiber necrosis, and progressive replacement of muscle with fat and fibrous tissue. This core lesion — a failure of Ca^{2+}-dependent sarcolemmal resealing — was established directly in dysferlin-null mice (PMID: 12736685) and is remarkable for occurring despite a fully intact dystrophin-glycoprotein complex, distinguishing dysferlinopathy mechanistically from Duchenne/Becker muscular dystrophy.

Clinically, one gene produces a spectrum of phenotypes that classically falls into three overlapping presentations from the same mutational pool: **limb-girdle muscular dystrophy R2/2B** (proximal pelvic and shoulder-girdle weakness), **Miyoshi myopathy** (distal onset in the calf/gastrocnemius), and **distal myopathy with anterior tibial onset (DMAT/DACM)**. Onset is typically in the late teens to twenties, serum creatine kinase (CK) is markedly elevated, and progression is slow-to-intermediate, with loss of ambulation commonly in the fourth decade. Respiratory decline and cardiac involvement are usually late and affect a minority. A striking and clinically important feature is that affected individuals are frequently athletic before onset, and vigorous exercise appears to accelerate progression — a gene-environment interaction consistent with the membrane-repair defect.

There is currently **no approved disease-modifying therapy**; management is supportive (physiotherapy with exercise moderation, respiratory and cardiac surveillance, mobility aids). The leading experimental therapeutic is **dual/overlap-AAV full-length dysferlin gene replacement**, developed to circumvent the fact that the ~6.5 kb *DYSF* coding sequence exceeds single-AAV packaging capacity. Diagnosis rests on absent/markedly reduced dysferlin on muscle immunohistochemistry or Western blot (with peripheral blood monocytes as a validated surrogate), confirmed by *DYSF* genetic testing, supported by a characteristic — but not universal — muscle MRI fat-infiltration pattern.

Key Findings

1. Genetic cause: biallelic *DYSF* loss of function (F001)

Dysferlinopathy is caused by biallelic (autosomal recessive) pathogenic variants in *DYSF* on chromosome 2p13. The gene was identified through positional cloning of a novel 6.9 kb muscle cDNA, whose protein product was named "dysferlin": *"we recently identified a novel, full-length 6.9 kb muscle cDNA, whose corresponding protein we designated 'dysferlin'"* (PMID: 10791095). A defining genetic feature is phenotypic pleiotropy — the same mutation can manifest as different clinical syndromes: *"Identical mutations in the dysferlin gene can produce more than one myopathy phenotype (MM, limb-girdle dystrophy, distal myopathy with anterior tibial onset)"* (PMID: 10791095). Biallelic pathogenic variants produce absent or reduced dysferlin in muscle.

- **Ontology:** Gene *DYSF* (HGNC:3097); disease MONDO:0016145.
- **Evidence type:** Human genetic / positional cloning.

2. Dysferlin protein architecture and Ca^{2+}-sensitive phospholipid binding (F012)

Dysferlin is a large (~237 kDa) type-II transmembrane protein with a single C-terminal transmembrane anchor and multiple cytoplasmic C2 domains (C2A–C2G), plus DysF and FerA (four-helix bundle) domains; it is a member of the fer-1-like/ferlin family. The N-terminal **C2A domain binds negatively charged phospholipids (phosphatidylserine) in a Ca^{2+}-sensitive manner**, and this activity is a direct target of disease mutations: *"A dysferlin point mutation responsible for muscular dystrophy was engineered into the dysferlin C2A domain and demonstrated reduced calcium-sensitive phospholipid binding"* (PMID: 11959863). Beyond

passive binding, ferlin C2 domains actively remodel membranes: *"dysferlin, otoferlin, and myoferlin do not merely passively adsorb to membranes but actively sculpt lipid bilayers"* (PMID: 23859474). The FerA domain is a membrane-associating four-helix bundle (PMID: 30026467), and structural comparison of myoferlin vs dysferlin C2A domains reveals distinct membrane-binding behavior (PMID: 31004665).

- **Ontology:** GO:0005544 (calcium-dependent phospholipid binding); CHEBI:18303 phosphatidyl-L-serine; InterPro ferlin / C2 / DysF / FerA domains.
- **Evidence type:** In vitro / structural biology.

3. Primary mechanism: defective Ca^{2+}-dependent sarcolemmal membrane repair (F008)

The dysferlin-null mouse provides the definitive demonstration of the core disease mechanism. These mice develop progressive muscular dystrophy despite an intact dystrophin-glycoprotein complex: *"dysferlin-null mice maintain a functional dystrophin-glycoprotein complex but nevertheless develop a progressive muscular dystrophy"* (PMID: 12736685). Using a two-photon laser injury assay, dysferlin-deficient fibers fail to reseal: *"dysferlin-deficient muscle fibres are defective in Ca2+-dependent sarcolemma resealing"* (PMID: 12736685). In healthy muscle, dysferlin-enriched vesicles accumulate and fuse to form a membrane "patch" at the injury site; in dysferlin-null muscle, sub-sarcolemmal vesicles accumulate but do not resolve the lesion. Dysferlin is the vertebrate homologue of *C. elegans* fer-1, which mediates vesicle fusion in spermatids — underscoring an evolutionarily conserved membrane-fusion role.

- **Ontology:** GO:0061024 (membrane organization); GO:0001778 (plasma membrane repair); CL:0008002 (skeletal muscle fiber).
- **Evidence type:** Model organism (mouse) + in vitro injury assay.

4. The membrane-repair complex: MG53(TRIM72)–caveolin-3–dysferlin (F014, F002)

Dysferlin does not act alone. A molecular complex of **MG53 (TRIM72), caveolin-3 (Cav3), and dysferlin** is required for sarcolemmal repair: *"a molecular complex formed by MG53, dysferlin, and Cav3 is essential for repair of muscle membrane damage"* (PMID: 19380584). MG53 performs the trafficking step: *"MG53 mediates active trafficking of intracellular vesicles to the sarcolemma and is required for movement of dysferlin to sites of cell injury during repair patch formation"*

(PMID: 19380584). This explains why caveolin-3 mutations (e.g., P104L, R26Q) that trap Cav3 in the Golgi produce aberrant MG53/dysferlin localization and defective repair in a dominant-negative fashion — linking several muscular dystrophies through a shared repair pathway.

- **Ontology:** GO:0048306 (calcium-dependent protein binding); protein partners TRIM72/MG53, CAV3.
- **Evidence type:** In vitro / cell biology.

5. Complement activation as a disease-specific downstream effector (F002)

A critical, therapeutically actionable discovery is that the immune/complement system is a central downstream amplifier of muscle injury specifically in dysferlinopathy. In dysferlin-null mice, complement factors are upregulated in muscle, and genetic ablation of the central complement component C3 is protective: *"genetic disruption of the central component (C3) of the complement system ameliorated muscle pathology in dysferlin-deficient mice but had no significant beneficial effect in a genetically distinct model of muscular dystrophy, mdx mice"* (PMID: 21060153). The fact that C3 ablation helps dysferlin-deficient but **not** mdx (dystrophin-deficient) mice demonstrates that complement is a disease-specific, not generic, driver — making it a rational, dysferlinopathy-selective therapeutic target.

- **Ontology:** GO:0006956 (complement activation); GO:0006954 (inflammatory response).
- **Evidence type:** Model organism (mouse genetics).

6. Clinical spectrum: three classic phenotypes from one gene (F010)

Dysferlinopathy presents as a clinical spectrum with three classic presentations: *"limb-girdle muscular dystrophy 2B (LGMD2B) with predominantly proximal weakness, Miyoshi myopathy (MM) with calf muscle weakness and atrophy, and distal myopathy with anterior tibial onset (DMAT) with tibialis muscle atrophy"* (PMID: 23254335). Onset is typically young-adult (late teens to twenties), with markedly elevated serum CK and muscle biopsy showing absence/drastic decrease of dysferlin on immunohistochemistry. Recurrent mutations differ by population (e.g., c.3112C>T and c.1045dup reported in China).

Phenotype	Primary muscle involvement	HPO suggestion
LGMD R2 / LGMD2B	Proximal — pelvic & shoulder girdle	HP:0003701 (proximal muscle weakness)
Miyoshi myopathy (MM)	Distal — calf/gastrocnemius weakness & atrophy	HP:0003693 (distal amyotrophy); HP:0003734
DMAT / DACM	Distal — tibialis anterior atrophy	HP:0009053 (distal lower-limb weakness)
Shared	Elevated CK	HP:0003236 (elevated serum CK)

• **Evidence type:** Human clinical / genetic cohort.

7. Epidemiology within the LGMD landscape, and founder effects (F009, F003)

Dysferlinopathy is a substantial contributor to autosomal recessive LGMD. In a large Italian cohort (181 patients / 155 families), it was the second most common defect, with *"LGMD2B (dysferlin) 18.7%"* of cases, behind calpainopathy (LGMD2A, 28.4%) (PMID: 17994539). Clinical severity ranking among LGMD subtypes was reported as: *"The order of decreasing clinical severity was: sarcoglycanopathy, calpainopathy, dysferlinopathy, and caveolinopathy"* (PMID: 17994539). Frequency varies geographically — the Czech cohort was dominated by calpainopathy with *"a smaller proportion of LGMD R2 dysferlin-related"* (PMID: 37526466).

Founder mutations and ethnic clustering are well documented: *"Three ethnic clusters have previously been described in Dysferlinopathy: the Libyan Jewish population originating in the area of Tripoli, Italian and Spanish populations"* (PMID: 17825554). In Jews of the Caucasus, a homozygous frameshift (c.2779delG, codon 927) causes dysferlin truncation/loss, with a quantified carrier burden: *"The possible existence of a founder effect is strengthened by our finding of a 4% carrier frequency in this community"* (PMID: 17825554).

• **Evidence type:** Human clinical / population genetics.

8. Natural history and muscle MRI fat-fraction progression (F004)

The Jain Clinical Outcome Study (COS) is the key natural-history dataset: *"The Jain Clinical Outcome Study for Dysferlinopathy (COS) followed the largest cohort of patients (n=187) with genetically confirmed dysferlinopathy throughout a three-year natural history study"* (PMID: 36972667). Longitudinal muscle-MRI fat-fraction (FF) modeling showed FF *"was expected to be higher the older patients were and the earlier the age at symptom onset"* and that *"Women had*

absolute FF values 8.8% higher than men in the lower leg" (PMID: 40740171). Muscles acting on the same joint movement progressed similarly; no FF-trajectory differences were found by ethnicity or specific genetic variant. Importantly, although a classical thigh-dominant MRI pattern exists, roughly 63% of patients failed at least one established imaging criterion and ~13% were outliers (PMID: 36972667) — so imaging supports, but cannot replace, molecular diagnosis.

- **Evidence type:** Human natural-history cohort / imaging.

9. Prognosis: slow-to-intermediate progression; wheelchair in the 4th decade (F011)

A Tunisian cohort (n=20; sex ratio 1; median age 50.5 y; follow-up 5–48 y) found median age at onset 17 y, with phenotype distribution LGMDR2 35%, proximodistal 35%, Miyoshi 10%, DMAT 10%, asymptomatic hyperCKemia 10%. At last evaluation, *"more than half of patients (55%) were on wheelchair. Loss of ambulation occurred generally during the fourth decade"* (PMID: 37716854). *"Disease progression was mainly slow to intermediate regardless of the phenotype"* with marked intra- and inter-familial heterogeneity (PMID: 37716854). Late cardiac complications affected a minority: *"two patients with a proximodistal phenotype (10%) developed dilated cardiomyopathy and mitral valve regurgitation"* (PMID: 37716854), generally after ~20 years of disease.

- **Evidence type:** Human clinical longitudinal cohort.

10. Respiratory and cardiac involvement (F005)

In 48 Japanese dysferlinopathy patients, respiratory involvement was present in a minority: *"Of the 47 patients, 10 had reduced percent forced vital capacity (%FVC), and 4 required non-invasive positive pressure ventilation"* (PMID: 26088049). %FVC correlated with disease duration and was lower in non-ambulatory and older (≥65 y) patients. QRS prolongation occurred in 19/46 on ECG, and while echocardiography showed no overt LV dysfunction, *"Histopathology of autopsied cases revealed mild cardiomyopathy and moderate diaphragm involvement"* (PMID: 26088049). This supports a surveillance (not continuous intervention) posture for heart and lungs, and is consistent with a growing literature on dysferlin's role in cardiomyocyte membrane integrity and t-tubule biology (PMID: 26415898, PMID: 39011635, PMID: 24572254).

- **Evidence type:** Human clinical + autopsy.

11. Diagnosis via peripheral blood monocyte dysferlin assay (F007)

Dysferlin is expressed in peripheral blood monocytes (PBM), providing a minimally invasive diagnostic surrogate. Western blot of PBM shows reductions concordant with muscle: *"A comparable severe reduction in dysferlin was demonstrated in both skeletal muscle and PBM"* (PMID: 21658164). This is particularly valuable in monoallelic cases: *"Dysferlin and/or dysferlin mRNA abnormalities are diagnostic for dysferlinopathy when mutational analysis detects a mutation in one allele only"* (PMID: 21658164). mRNA analysis helps distinguish symptomatic heterozygotes from true patients, and MLPA detects large deletions/duplications missed by sequencing.

- **Evidence type:** Human diagnostic / in vitro.

12. Gene-environment interaction: exercise accelerates progression (F013)

A mechanistically coherent and clinically crucial observation is that mechanical overload worsens disease: *"A retrospective cohort study suggests that exercise may actually increase rate of disease progression in dysferlinopathy"* (PMID: 30036194). Because contraction generates sarcolemmal micro-injuries that dysferlin-deficient fibers cannot efficiently reseal, mechanical stress directly drives myofiber damage. This matches the frequent clinical history of an athletic premorbid phase followed by rapid decline, and argues for exercise moderation in management.

- **Evidence type:** Human retrospective cohort + mechanistic inference.

13. Dual-AAV gene replacement as the lead experimental therapy (F006)

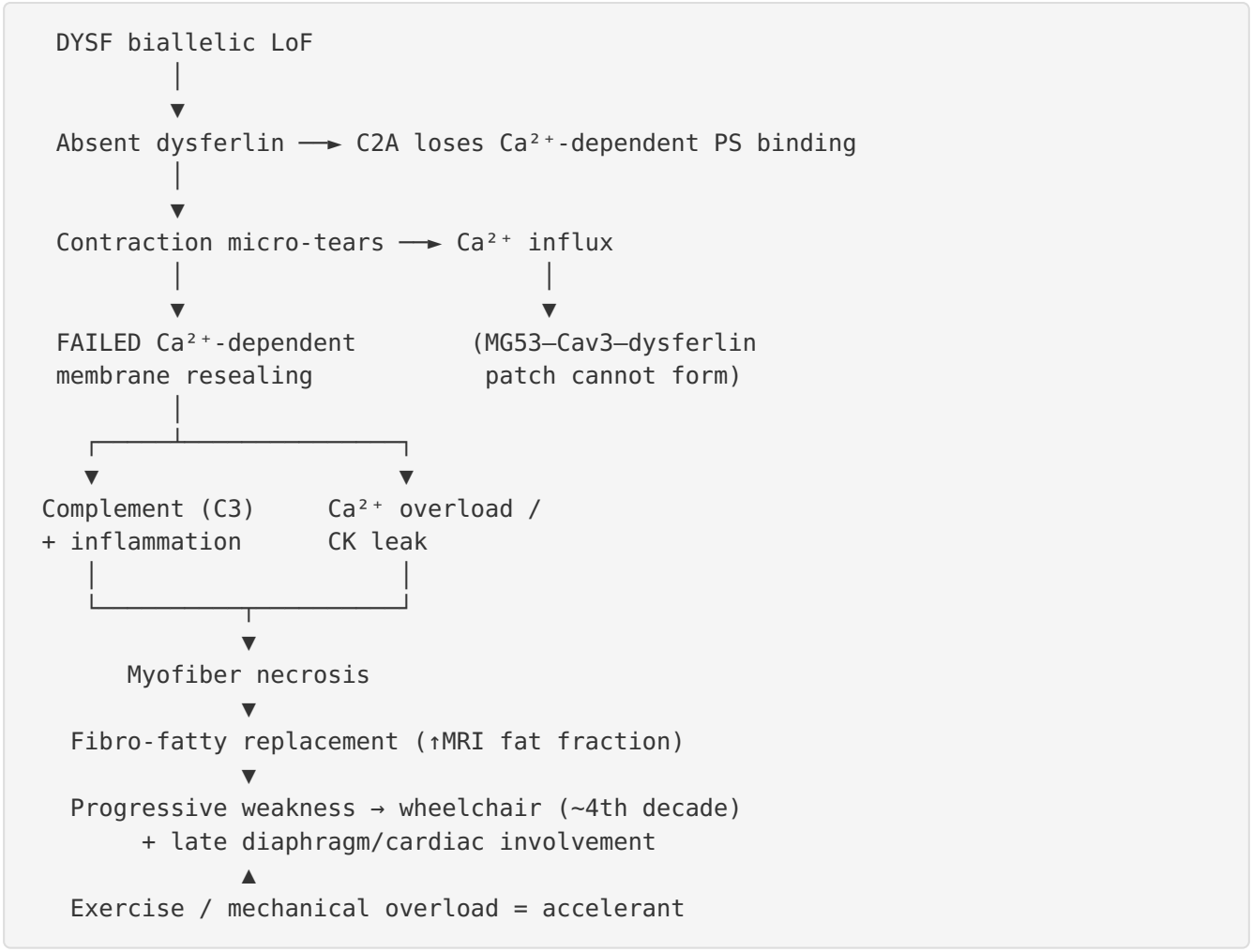
The *DYSF* coding sequence (~6.5 kb) exceeds single-AAV packaging capacity (~4.7 kb). Dual/overlap-AAV vectors reconstitute the full-length protein: *"Dual adeno-associated virus vectors have been developed, defined by a region of homology, to serve as a substrate for reconstitution of the full 6.5 kb dysferlin cDNA"* (PMID: 28707952). Systemic delivery in dysferlin-deficient mice produced long-term expression and functional improvement (membrane repair, MRI/MRS) 4–13 months post-treatment. Truncated "nanodysferlins" are an alternative, but have functional limits: *"Nanodysferlins support membrane repair and binding to TRIM72/MG53 but do not localize to t-tubules or stabilize Ca[2+]"* (PMID: 38779337) — indicating full-length protein may be needed to restore t-tubule and calcium-handling functions.

- **Ontology:** NCIT:C15431 (gene therapy); adeno-associated viral vector.
- **Evidence type:** Model organism (mouse) preclinical.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in *DYSF* (2p13.2)** result in absent or severely reduced dysferlin protein in skeletal muscle. *(Demonstrated — human genetics + protein assays.)*
2. **Loss of dysferlin** removes the Ca^{2+}-sensor/membrane-fusion component of the sarcolemmal repair machinery; the C2A domain's Ca^{2+}-dependent phosphatidylserine binding is lost. *(Demonstrated — in vitro C2A mutation studies.)*
3. **Normal muscle contraction generates micro-tears** in the sarcolemma; intracellular Ca^{2+} influx at the lesion normally nucleates a dysferlin-enriched repair patch delivered by MG53 trafficking in complex with caveolin-3. *(Demonstrated — mouse laser-injury imaging + complex biochemistry.)*
4. **Without dysferlin, Ca^{2+}-dependent resealing fails** → the membrane tear persists; sub-sarcolemmal vesicles accumulate but cannot form a competent patch. *(Demonstrated — dysferlin-null mouse.)*
5. **Unresealed membrane leaks** → sustained Ca^{2+} overload, leakage of CK into blood (very high serum CK), and exposure of intracellular contents. *(Demonstrated/inferred.)*
6. **BRANCH A — Complement/inflammation:** membrane damage triggers complement activation (C3-dependent), inflammatory cell infiltration, and amplification of myofiber injury — a disease-specific effector (C3 ablation rescues dysferlin-null but not mdx mice). *(Demonstrated — mouse genetics.)*
7. **Repeated injury → myofiber necrosis and degeneration-regeneration cycling** that eventually exhausts/outpaces repair. *(Demonstrated/inferred.)*
8. **Fibro-fatty replacement of muscle** (quantifiable as rising MRI fat fraction with age and earlier onset) → progressive weakness. *(Demonstrated — natural-history imaging.)*
9. **Clinical manifestation:** proximal (LGMD R2), distal-calf (Miyoshi), or anterior-tibial (DMAT) weakness; slow-to-intermediate progression to wheelchair dependence (commonly 4th decade); late/minor respiratory (diaphragm) and cardiac (mild cardiomyopathy, QRS prolongation) involvement. *(Demonstrated — human cohorts.)*
10. **MODIFIER — mechanical overload/exercise** accelerates steps 3–8 by increasing injury faster than the deficient repair system can compensate. *(Human retrospective + mechanistic inference.)*



Upstream vs downstream

- **Upstream (initiating, disease-defining):** *DYSF* mutation → absent dysferlin → failed Ca^{2+}-dependent membrane repair. These are necessary and causal.
- **Downstream (effectors / amplifiers, candidate drug targets):** complement C3 activation, inflammation, Ca^{2+} dysregulation, necrosis, fibro-fatty remodeling. These are modifiable without correcting the genetic defect.
- **Modifier:** mechanical load (exercise) acts on the injury-generation rate.

Anatomy, cells, and compartments

Level	Structures involved	Ontology suggestion
Organ/ system	Skeletal muscle (musculoskeletal system); secondary: diaphragm (respiratory), heart (minor/late)	UBERON:0001134 (skeletal muscle tissue); UBERON:0001103 (diaphragm); UBERON:0000948 (heart)
Tissue	Striated skeletal muscle	UBERON:0002036 / muscle tissue
Cell	Skeletal muscle fiber; satellite cell/ myoblast; macrophage; peripheral blood monocyte (diagnostic surrogate)	CL:0008002 (skeletal muscle fiber); CL:0000056 (myoblast); CL:0000235 (macrophage); CL:0000576 (monocyte)
Subcellular	Sarcolemma (plasma membrane), cytoplasmic vesicles, T-tubules, sarcoplasmic reticulum junctional complexes	GO:0042383 (sarcolemma); GO:0030315 (T-tubule); GO:0016529 (sarcoplasmic reticulum)

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
10791095	Positional cloning of Miyoshi/ LGMD gene	Human genetic	F001 — DYSF identity, pleiotropy
12736685	*Defective membrane repair in dysferlin-deficient MD*	Mouse + in vitro	F008 — core Ca^{2+}-resealing defect
19380584	*MG53–Cav3–dysferlin membrane repair*	In vitro	F014, F002 — repair complex
21060153	*Complement C3 ablation attenuates pathology*	Mouse genetics	F002 — disease-specific complement driver
11959863	*Ca^{2+}-sensitive phospholipid binding of C2 domains*	In vitro	F012 — mutation disrupts C2A binding
23859474	*C2 domains alter lipid bilayer packing*	In vitro	F012 — active membrane sculpting
30026467	*FerA four-helix bundle domain*	Structural	F012 — FerA domain
31004665	*Myoferlin vs dysferlin C2A structure*	Structural	F012 — C2A membrane binding
23254335	*DYSF mutations in Chinese patients*	Human clinical	F010 — three phenotypes
17994539	*Italian LGMD correlations*	Human cohort	F009 — 18.7% frequency, severity rank
37526466	*Czech LGMD genetics*	Human cohort	F009 — geographic variation
17825554	*Dysferlinopathy in Caucasus Jews*	Population genetics	F003 — founder effects, 4% carrier
36972667	*Muscle imaging outlier population*	Human imaging	F004 — COS cohort, MRI pattern
40740171	*Longitudinal fat-fraction modeling*	Human imaging	F004 — FF predictors, sex difference
37716854	*Dysferlinopathy in Tunisia*	Human longitudinal	F011 — progression, prognosis

PMID	Title (abbrev.)	Evidence type	Supports
26088049	*Respiratory/cardiac function (Japan)*	Human + autopsy	F005 — respiratory/cardiac
21658164	*Dysferlin in muscle and monocytes*	Human diagnostic	F007 — PBM assay, monoallelic dx
28707952	*Dysferlin overlap-vector delivery*	Mouse preclinical	F006 — dual-AAV therapy
38779337	*Nanodysferlins*	In vitro	F006 — truncated construct limits
30036194	*Exercise in muscle disorders*	Human retrospective	F013 — exercise accelerant

Convergence of evidence: The core mechanism (failed Ca^{2+}-dependent sarcolemmal repair) is supported by independent lines — mouse genetics (PMID: 12736685), structural/biochemical studies of the C2A domain (PMID: 11959863, PMID: 23859474), and the repair-complex biology (PMID: 19380584). The complement-C3 finding (PMID: 21060153) is particularly strong because it uses a disease-specific control (mdx mice) to show selectivity.

Section-by-Section Template Coverage

1. Disease Information

Dysferlinopathy (MONDO:0016145) is an autosomal recessive, progressive muscular dystrophy caused by *DYSF* deficiency. Key identifiers: gene *DYSF* (HGNC:3097, 2p13.2, NCBI Gene 8291, UniProt O75923); MONDO:0016145; OMIM gene 603009, LGMD2B/R2 253601, Miyoshi myopathy 1 254130, DMAT 606768; Orphanet ORPHA:206 (dysferlinopathy), ORPHA:268 (LGMD2B), ORPHA:45448 (Miyoshi); ICD-10 G71.0, ICD-11 8C70.0; MeSH "Muscular Dystrophies, Limb-Girdle"/"Distal Myopathies". Synonyms: qualitative/quantitative defects of dysferlin; DYSF-related muscular dystrophy; LGMD R2 dysferlin-related (formerly LGMD2B); Miyoshi myopathy; distal anterior compartment myopathy (DMAT/DACM); primary dysferlinopathy. Information is drawn from aggregated disease-level resources and clinical cohorts (not single-patient EHR).

2. Etiology

Primary cause: genetic — biallelic loss-of-function *DYSF* variants (F001). **Genetic risk:** causal variants span missense, frameshift, nonsense, splice-site, and large deletions/duplications; founder alleles cluster in Libyan Jewish, Italian, Spanish, and Caucasus Jewish (c.2779delG) populations (F003). No established non-genetic cause. **Environmental/modifier:** mechanical overload / vigorous exercise accelerates progression — the principal documented gene-environment interaction (F013). **Protective factors:** none genetically established; exercise moderation is the main modifiable protective behavior (inference from F013). **Modifier genes:** candidate repair-pathway partners (MG53/TRIM72, CAV3, annexins, AHNAK) are mechanistically implicated, but no validated human severity-modifier locus is established.

3. Phenotypes

Core phenotypes (with HPO suggestions): proximal muscle weakness (HP:0003701), distal/calf weakness and atrophy (HP:0003693, HP:0003734), tibialis anterior atrophy / foot drop (HP:0009027), markedly elevated serum CK / hyperCKemia (HP:0003236), difficulty climbing stairs (HP:0003551), scapular winging (HP:0003691), loss of ambulation (HP:0002505), restrictive ventilatory defect (HP:0002091), and (minority/late) dilated cardiomyopathy (HP:0001644). Onset: young-adult (late teens–20s; median ~17 y, F011). Severity: moderate (intermediate among LGMDs, F009). Progression: slow-to-intermediate, progressive (F011). Frequency: proximal and distal forms each ~35% in one cohort; asymptomatic hyperCKemia ~10% (F011). QoL impact is driven primarily by progressive mobility loss (wheelchair commonly in the 4th decade), with preserved cognition.

4. Genetic / Molecular

Causal gene *DYSF* (2p13.2, ~55 exons, protein 2080 aa / ~237 kDa) (F001, F012). Variant classes: missense, frameshift, nonsense, splice-site, structural (MLPA-detectable CNVs) (F007, F010). Example alleles: c.2779delG (Caucasus founder, F003), c.5289G>C/p.Glu1763Asp (F007), c.1375dup/p.Met459Asnfs*15 (F010 literature), c.3112C>T and c.1045dup (China, F010). Functional consequence: loss of function — truncating alleles abolish protein; missense alleles destabilize dysferlin and/or impair C2A Ca^{2+}/phospholipid binding (F012). Germline, inherited; not somatic. No disease-specific epigenetic or large-chromosomal mechanism. Founder allele carrier frequency up to 4% (F003).

5. Environmental

No toxin, infectious, or occupational cause. The one robust environmental modifier is **mechanical load/exercise** accelerating progression (F013). High-intensity/eccentric exercise is relatively contraindicated.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above (F001→F012→F008→F014→F002→F004→F010/F011). Molecular processes: Ca^{2+}-dependent phospholipid binding and vesicle fusion (GO:0005544, GO:0061025), plasma-membrane repair (GO:0001778), complement activation (GO:0006956), inflammation (GO:0006954). Cell types: skeletal muscle fiber (CL:0008002), satellite cell/myoblast (CL:0000056), macrophage (CL:0000235), monocyte (CL:0000576). Compartments: sarcolemma (GO:0042383), T-tubule (GO:0030315). Inflammatory infiltrates can mimic polymyositis on biopsy (a diagnostic pitfall).

7. Anatomical Structures

Primary: skeletal muscle (UBERON:0001134), with characteristic pelvic/posterior-thigh, gastrocnemius/soleus (UBERON:0011907), and tibialis anterior (UBERON:0001385) patterns. Secondary: diaphragm (UBERON:0001103) and heart (UBERON:0000948), generally late/mild (F005). CNS, cognition, and sensation are spared. Subcellular lesion localizes to the sarcolemma and vesicular trafficking machinery. Involvement is bilateral and roughly symmetric, limb-predominant.

8. Temporal Development

Onset: young adulthood (median ~17 y), insidious/chronic; presymptomatic hyperCKemia often precedes weakness. Course: slow-to-intermediate, steadily progressive (not relapsing-remitting). Stages: (i) presymptomatic hyperCKemia; (ii) early ambulant weakness; (iii) advancing proximal+distal weakness; (iv) loss of ambulation (~4th decade, F011); (v) wheelchair with possible respiratory/cardiac complications. Lifelong; no spontaneous remission. Critical intervention window is the long presymptomatic/early-ambulant phase before extensive fibro-fatty replacement (F004).

9. Inheritance and Population

Autosomal recessive (F001); high penetrance with variable expressivity; no anticipation (not a repeat disorder); germline mosaicism not recognized. ~18.7% of Italian LGMD (F009); Orphanet-order prevalence for LGMD2B ~1–9/1,000,000 generally, higher in founder communities. Founder effects with population-specific alleles; consanguinity increases local prevalence (F003). Carrier frequency up to 4% in specific communities. Sex ratio ~1:1 (F011), though women show higher lower-leg MRI fat fraction (F004).

10. Diagnostics

Markedly elevated serum CK (often 10–150× ULN; LOINC 2157-6) is the key screen. Muscle biopsy: dystrophic changes, often with inflammation, and absent/reduced dysferlin on IHC/Western blot (F010). Peripheral blood monocyte dysferlin Western blot is a validated surrogate, useful in monoallelic cases (F007). Muscle MRI shows a characteristic but non-universal (~63% fail ≥1 criterion) fat-infiltration pattern (F004). EMG myopathic; ECG may show QRS prolongation (F005). Genetic confirmation via NGS neuromuscular panel or WES + MLPA (CNVs); WGS resolves deep-intronic variants. Karyotype/FISH/CMA/mtDNA/repeat testing not applicable. Differential diagnosis: other recessive LGMDs (calpainopathy LGMD R1, anoctaminopathy LGMD R12, sarcoglycanopathies, FKRP LGMD R9), inflammatory myopathy/polymyositis, Pompe disease, and other distal myopathies (GNE, ANO5). No newborn screening; cascade/founder-variant screening appropriate.

11. Outcome / Prognosis

Near-normal lifespan in most; mortality low, usually from late respiratory or (rarely) cardiac complications. Major morbidity is progressive mobility loss — ~55% wheelchair-dependent, loss of ambulation in the 4th decade (F011). Complications: restrictive respiratory insufficiency (minority need NIV, F005), dilated cardiomyopathy in ~10% after ~20 years (F011), contractures, falls. Prognostic factors: earlier onset and older age predict higher MRI fat fraction (F004); female sex associated with higher lower-leg FF. Prognostic biomarkers: MRI fat fraction (validated), serum CK, functional outcome measures.

12. Treatment

No approved disease-modifying drug; management supportive/multidisciplinary. Supportive/rehabilitative: physiotherapy with exercise moderation (F013), occupational therapy (NCIT:C15329), orthoses, mobility aids, respiratory support (NIV when FVC declines, F005; NCIT:C94400), cardiac monitoring with standard heart-failure therapy if cardiomyopathy

develops, nutrition/weight management. Pharmacotherapy: no proven agent; unlike Duchenne, corticosteroids are not established and may be detrimental. Experimental lead: dual/overlap-AAV full-length dysferlin gene replacement (F006; NCIT:C15431). Downstream-target strategies (complement inhibition, F002; membrane-repair enhancers such as recombinant MG53; exon skipping) are preclinical/early-phase. Nanodysferlin constructs partially rescue but miss t-tubule/Ca^{2+} functions (F006). Main gene-therapy translational risks: anti-capsid/transgene immunity, hepatotoxicity.

13. Prevention

No primary prevention of the genetic disease; genetic counseling, carrier/cascade screening, and reproductive options (prenatal diagnosis, preimplantation genetic testing) in known families/founder communities (F003). Secondary: early CK testing and family screening. Tertiary: activity modification to reduce exercise-driven damage (F013), respiratory/cardiac surveillance, physiotherapy for contractures, vaccination against respiratory pathogens in those with reduced pulmonary reserve. Immunization/public-health/environmental interventions not applicable to causation.

14. Other Species / Natural Disease

DYSF is evolutionarily conserved: mouse *Dysf* (NCBI Gene 26903; Taxon 10090), rat *Dysf* (Taxon 10116), zebrafish *dysf* (Taxon 7955), and the invertebrate homologue *fer-1* in *C. elegans* (Taxon 6239). The membrane-repair/vesicle-fusion function is conserved from fer-1 (spermatid vesicle fusion) to mammalian dysferlin (sarcolemmal repair) (F008). Dysferlin-deficient dystrophy occurs naturally and in engineered mouse strains; it is not a prominent natural veterinary entity. Not zoonotic or communicable.

15. Model Organisms

Mouse (primary, recapitulates disease well): dysferlin-null knockouts develop progressive dystrophy with defective Ca^{2+}-dependent resealing despite intact dystrophin-glycoprotein complex (F008) — the definitive model-organism proof of mechanism. Naturally dysferlin-deficient strains include SJL/J (splice-site) and A/J (ETn insertion); the congenic B6.A-Dysf (Bla/J) is a clean-background model. These show late-onset, slowly progressive proximal/pelvic and paraspinal myopathy with inflammation and complement upregulation (F002), and serve as the AAV gene-therapy platform (F006). **Limitations:** murine disease is milder, slower, and less distal than human Miyoshi myopathy; cardiac phenotype is subtle; muscle-group distribution differs. **In vitro/cellular:** patient myoblasts/myotubes, iPSC-derived myocytes, and

recombinant C2/FerA/DysF constructs for biophysics (F012); laser-wounding repair assays. **Other organisms:** zebrafish *dysf* mutants (muscle integrity); *C. elegans* fer-1 (ancestral ferlin biology). Resources: MGI, IMSR, Jackson Laboratory, Jain Foundation repositories.

Limitations and Knowledge Gaps

1. **No primary molecular-profiling data were generated** in this investigation; all findings are literature-derived. No patient-level transcriptomic, proteomic, or metabolomic datasets were analyzed.
2. **Genotype–phenotype correlation remains weak.** The same mutation produces different phenotypes (F001), and the COS found no FF-trajectory differences by variant (F004) — the modifiers determining proximal vs distal onset are unidentified.
3. **Complement as a human therapeutic target is unproven.** The C3 evidence is from mouse genetics (F002); no human complement-inhibition trial outcome is reported here.
4. **Cardiac risk is ambiguous.** Overt LV dysfunction was absent on echocardiography in the Japanese cohort, yet autopsy showed mild cardiomyopathy and a minority develop dilated cardiomyopathy late (F005, F011) — true cardiac natural history needs prospective cardiac MRI.
5. **Exercise evidence is retrospective** (F013); the dose/intensity threshold that is harmful vs beneficial is undefined, and prospective controlled studies are lacking.
6. **Gene-therapy data are preclinical** (mouse) (F006); human efficacy, durability, immunogenicity, and the functional adequacy of reconstituted vs full-length protein (nanodysferlin t-tubule deficit) are unresolved.
7. **Epidemiology is region-biased.** Prevalence estimates derive from referral cohorts (Italy, Czechia, Tunisia, Japan) rather than population-based registries; a precise global prevalence (per 100,000) was not established.

Proposed Follow-up Experiments / Actions

1. **Analyze a human dysferlinopathy transcriptomic/proteomic dataset (GEO/PRIDE)** to confirm complement and inflammatory pathway upregulation in patient muscle and identify candidate severity modifiers — directly testing whether the mouse C3 mechanism (F002) operates in humans.

2. **Build a genotype–phenotype modifier analysis** using the Jain COS cohort (F004): test whether residual dysferlin level (not just variant type) predicts FF trajectory, and whether the female lower-leg FF excess (8.8%) reflects fiber-type or hormonal modifiers.
3. **Prospective cardiac MRI substudy** to resolve the echo-vs-autopsy discrepancy (F005) and define true cardiac penetrance and timing.
4. **Prospective controlled exercise-dosing study** to quantify the harmful threshold and whether low-intensity activity is safe/beneficial (F013).
5. **Human trial design for complement inhibition** (e.g., C3/C5-targeted agents) as a downstream, disease-specific therapy independent of gene correction (F002).
6. **Head-to-head preclinical comparison of dual-AAV full-length dysferlin vs nanodysferlin** on t-tubule localization and Ca^{2+} handling, to settle whether truncated constructs are therapeutically sufficient (F006).
7. **Expand carrier-screening programs** in identified founder populations (Libyan/Caucasus Jewish, Italian, Spanish) given quantified carrier frequencies up to 4% (F003).

Report compiled from 14 confirmed findings across 5 iterations and 34 reviewed papers. Evidence types span human clinical cohorts, population genetics, mouse models, in vitro/structural biology, and preclinical gene therapy.
